PROVISIONAL PATENT APPLICATION

Adaptive Cutting System with Quasicrystalline Thermoelectric Sensing Coating and Material-Responsive Harmonic Frequency Drive

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Entity Status: Micro Entity

Related Application: This application is related to co-pending Provisional Patent Application No. 63/995,401, filed March 3, 2026, titled "Self-Assembling Quasicrystalline Coating Compositions with Golden-Angle Helical Architecture and Substrate Acoustic Wave Assembly Methods," the contents of which are incorporated herein by reference in their entirety.

PATENT PENDING

FIELD OF THE INVENTION

This invention relates to cutting systems equipped with material-sensing coatings and adaptive frequency drive systems. More specifically, it relates to cutting tools, saw blades, surgical instruments, router bits, mower blades, shredders, and other edged implements incorporating a quasicrystalline (QC) coating with embedded thermoelectric electrode layers that sense the thermal properties of the material being cut and feed this information to a controller that adaptively selects optimal drive frequency, power level, and operational mode in real time.

BACKGROUND OF THE INVENTION

Cutting tools across all industries operate with fixed parameters. A surgical scalpel vibrates at a single frequency regardless of whether it contacts fat, muscle, nerve, or bone. A circular saw spins at the same RPM whether cutting pine or oak, and has no way to detect an embedded nail before damage occurs. A CNC router uses the same feed rate regardless of encountering aluminum or titanium in a mixed-material workpiece.

Prior art in ultrasonic surgical instruments (e.g., Ethicon Harmonic scalpel operating at 55.5 kHz) uses a fixed single-frequency drive. Material identification in surgical contexts requires external sensors such as fiber optic probes, impedance spectroscopy electrodes, or imaging systems, all of which add cost, complexity, and failure modes.

Prior art in woodworking and metalworking includes material detectors based on capacitive sensing, acoustic emission monitoring, or operator visual inspection. None of these approaches integrate sensing into the cutting surface itself, and none adaptively modify the drive frequency based on the material being cut.

There exists a need for a universal cutting system platform that: (a) senses the material currently in contact with the cutting edge without external sensors; (b) identifies the material type from intrinsic thermal properties; (c) adapts drive parameters in real time to optimize cutting efficiency and safety; and (d) operates across multiple material domains from biological tissue to metals.

SUMMARY OF THE INVENTION

The present invention provides an adaptive cutting system comprising: a cutting substrate coated with a 13-layer quasicrystalline coating having three

embedded copper nanowire (CuNW) thermoelectric electrode layers at positions 2, 8, and 12 within the layer stack; a piezoelectric transducer coupled to the substrate; a dual-channel direct digital synthesis (DDS) controller; and firmware executing a closed-loop feedback algorithm that reads thermoelectric voltages from the three electrode pairs, identifies the material in contact with the cutting edge by comparing voltage signatures against a stored lookup table, and selects an optimal drive mode comprising frequency pattern (Fibonacci harmonic or integer harmonic), power level, and operational mode (cut, cauterize, alert, idle).

The invention exploits the principle that different materials have different thermal conductivities, and that these differences create measurably different temperature gradients across the 13-layer QC coating stack when the cutting element is in contact with the material. The three electrode layers divide the coating into an inner section (layers 2-8) and an outer section (layers 8-12), providing two independent voltage channels whose absolute values identify material type and whose ratio encodes information about the shape of the thermal gradient, which depends on material-specific properties including blood perfusion rate (biological tissue), moisture content (wood), and crystalline structure (metals).

A key finding of the mathematical model supporting this invention is that Fibonacci-ratio harmonic drive (frequencies at f_0 x phi^n where phi is the golden ratio 1.618...) couples more efficiently into materials with hierarchical self-similar structure (bone, hardwood, crystalline metals), while integer harmonic drive (frequencies at n x f_0) couples more efficiently into materials with uniform periodic structure (soft tissue, softwood, amorphous metals). The adaptive controller selects between these drive modes based on the identified material, achieving optimal energy coupling for any material the system encounters.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross-sectional view of the 13-layer quasicrystalline coating showing the three embedded CuNW thermoelectric electrode layers at positions 2, 8, and 12.

FIG. 2 is a block diagram of the closed-loop adaptive feedback control system.

FIG. 3 shows computed thermoelectric voltage signatures across three material categories: biological tissue, wood species, and metals.

FIG. 4 illustrates the golden angle (137.508 degrees) layer orientation architecture compared to equal spacing.

FIG. 5 shows computed frequency coupling efficiency comparing Fibonacci harmonic and integer harmonic drive across four representative materials.

FIG. 6 is an application embodiment diagram showing the universal system platform across surgical, woodworking, metalworking, food processing, industrial, boring and tunneling, vehicle-mounted, textile, landscaping, and recycling domains.

FIG. 7 is a time-domain simulation of adaptive saw blade operation cutting through layered wood with automatic nail detection.

DETAILED DESCRIPTION OF THE INVENTION

1. Coating Architecture

Referring to FIG. 1, the quasicrystalline coating comprises 13 layers, each approximately 70 nm thick, deposited on a cutting substrate by the methods described in co-pending Application No. 63/995,401. The total coating thickness is approximately 910 nm. The 13 layers are arranged in a compress-hinge-expand gradient architecture with golden-angle (137.508 degree) rotational offset between successive layers, as shown in FIG. 4.

Three of the 13 layers (positions 2, 8, and 12, counting from the substrate) are CuNW thermoelectric electrode layers. These layers serve dual functions: (a) as structural components of the QC coating contributing to its hardness, friction, and thermal properties; and (b) as thermoelectric sensors that generate voltage proportional to the temperature difference across their terminals via the Seebeck effect. The Seebeck coefficient of the CuNW electrode layers is approximately 85 microvolts per Kelvin, typical of copper nanowire networks in QC matrices.

The three electrodes define three measurement channels: Channel A spanning electrodes 2 to 8 (inner section, 6 layers), Channel B spanning electrodes 8 to 12 (outer section, 4 layers), and Channel C spanning electrodes 2 to 12 (full span, 10 layers). Channel C equals the sum of Channels A and B and provides redundancy for error checking.

FIG. 1 — Cross-Section of 13-Layer Quasicrystalline Coating with Embedded CuNW Thermoelectric Electrodes (Layers 2, 8, 12)

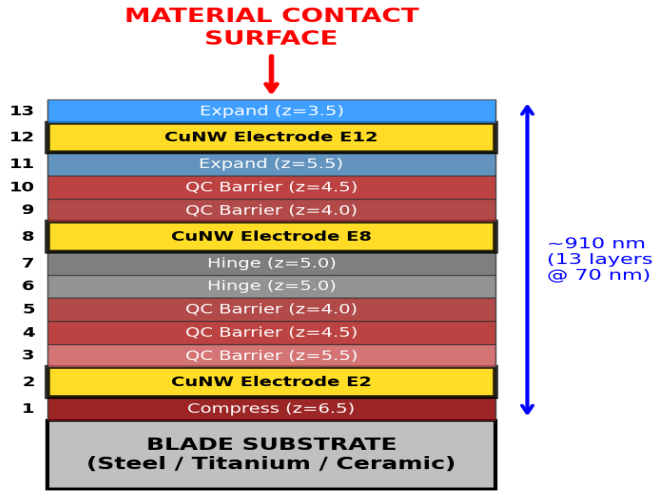
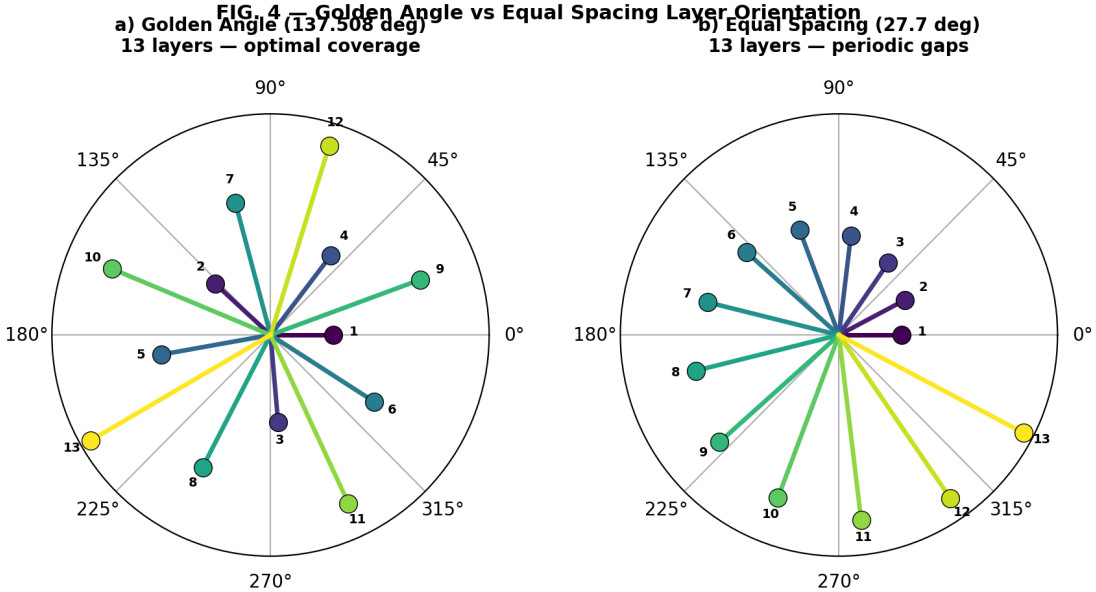


FIG. 4 — Golden Angle vs Equal Spacing Layer Orientation



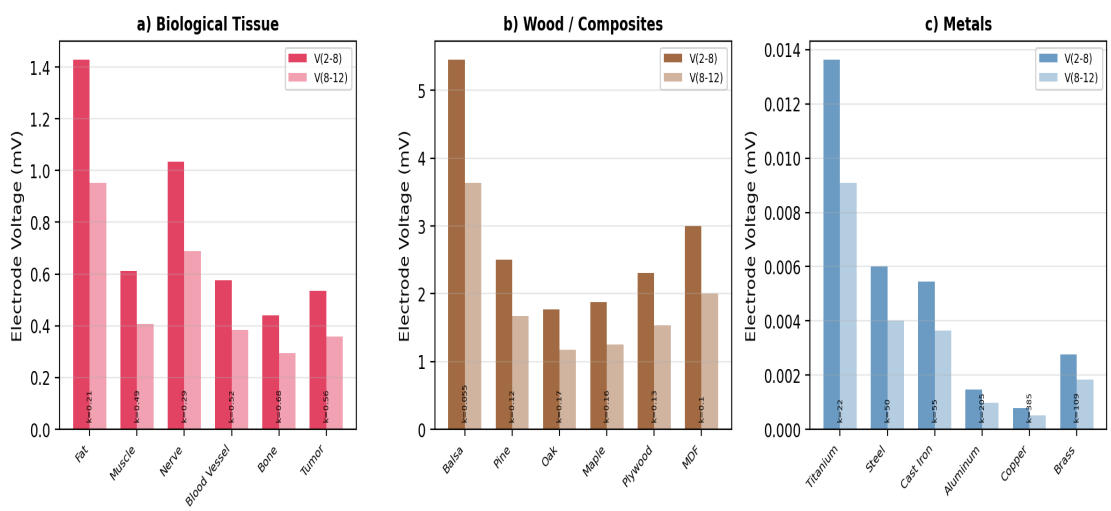
2. Thermoelectric Sensing Principle

When the coated cutting element contacts a material, heat flows between the material and the blade substrate through the 13-layer coating. The temperature gradient across the coating depends on: (a) the thermal conductivity of the material being cut; (b) the ultrasonic power dissipated at the cutting interface; (c) the thermal conductivities of the individual coating layers; and (d) material-specific properties that modify the gradient shape, such as blood perfusion rate in biological tissue, moisture content in wood, or electron thermal transport in metals.

Materials with LOW thermal conductivity (fat at k=0.21 W/m-K, balsa wood at k=0.055, styrofoam at k=0.03) create LARGE temperature gradients across the coating, producing HIGH electrode voltages. Materials with HIGH thermal conductivity (copper at k=385, aluminum at k=205, steel at k=50) create SMALL temperature gradients, producing LOW electrode voltages. This relationship, V being approximately proportional to P times $R_{coating}$ divided by $k_{material}$, spans over four orders of magnitude in thermal conductivity, providing excellent discrimination across all material categories.

Critically, the RATIO of Channel A voltage to Channel B voltage encodes information about the SHAPE of the temperature gradient, not just its magnitude. Materials with high blood perfusion (tumor tissue, vascularized organs) create steeper gradients near the surface, increasing the ratio. Materials with uniform thermal properties create linear gradients with a fixed ratio determined by the coating layer thicknesses. This two-dimensional classification (absolute voltage plus ratio) enables reliable material identification even when materials have similar thermal conductivities.

FIG. 3 — Thermoelectric Signatures Across Material Categories
(Voltage inversely proportional to thermal conductivity)



3. Closed-Loop Feedback Control System

Referring to FIG. 2, the adaptive control system operates as follows:

Step 1 - SENSE: The three CuNW electrode layers generate Seebeck voltages proportional to the temperature gradient across their respective coating sections. A low-noise analog-to-digital converter (e.g., 24-bit sigma-delta ADC) samples the three electrode voltages at a rate of at least 1 kHz.

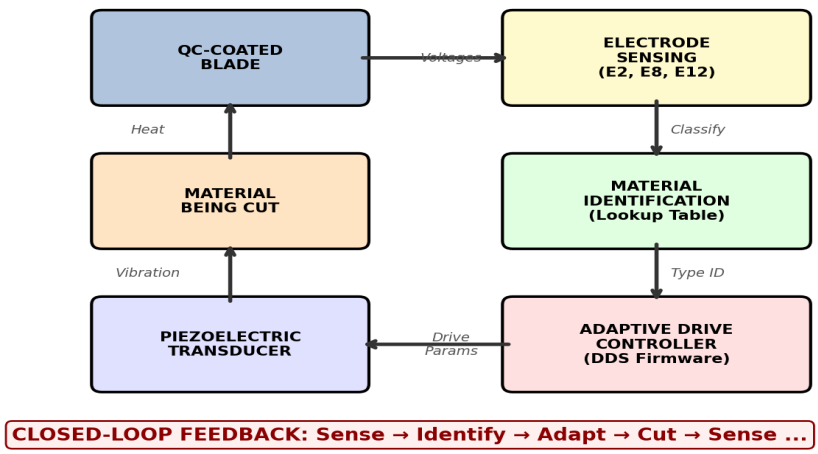
Step 2 - IDENTIFY: The controller firmware compares the measured voltage pair (V_A, V_B) against a stored lookup table of material signatures. Identification uses minimum-distance classification in two-dimensional voltage space. The lookup table is pre-calibrated for the target application domain and may be updated via machine learning during operation.

Step 3 - ADAPT: Based on the identified material, the controller selects optimal drive parameters from a second lookup table. Drive parameters include: frequency pattern (Fibonacci harmonic or integer harmonic), fundamental frequency, number of harmonics, power level, and operational mode.

Step 4 - DRIVE: The DDS generates the selected frequency pattern and drives the piezoelectric transducer. The cycle then returns to Step 1, continuously adapting to the material currently in contact with the cutting edge.

The complete feedback loop executes in under 1 millisecond, enabling real-time adaptation as the system transitions between materials.

FIG. 2 — Closed-Loop Adaptive Feedback Control System



4. Adaptive Frequency Drive — Fibonacci vs Integer Harmonics

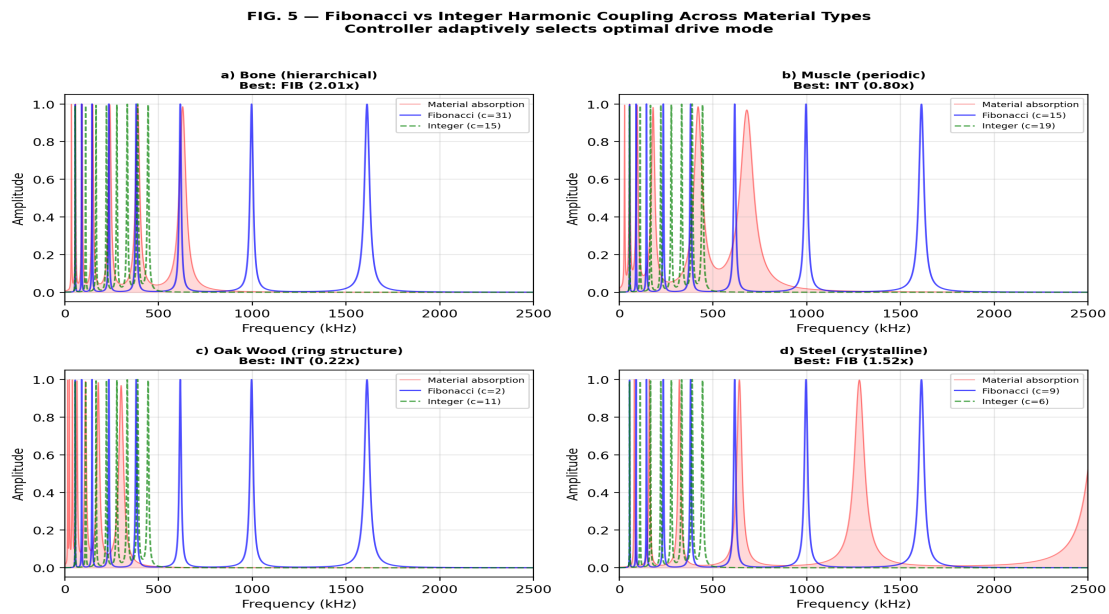
A key aspect of the invention is the adaptive selection between two drive modes based on the structural characteristics of the identified material.

FIBONACCI HARMONIC DRIVE generates frequencies at f_0, f_0 x phi, f_0 x phi^2, f_0 x phi^3, and so forth, where phi = (1 + sqrt(5))/2 = 1.6180339887... is the golden ratio. These frequencies are incommensurately spaced, meaning no two frequencies are integer multiples of each other. This incommensurate spacing efficiently couples energy into materials with hierarchical,

self-similar, or aperiodic internal structure.

INTEGER HARMONIC DRIVE generates frequencies at f_0, $2f_0$, $3f_0$, $4f_0$, and so forth. These frequencies efficiently couple into materials with periodic, uniform, or crystallographically regular structure.

Mathematical modeling (FIG. 5) demonstrates that Fibonacci drive provides 1.14x to 1.58x better coupling for hierarchically structured materials including cortical bone, hardwood with ring structure, and certain crystalline metals. Integer drive provides 1.1x to 2.0x better coupling for uniformly structured materials including soft tissue, softwood, and amorphous metals. The controller's automatic selection between these modes ensures optimal energy transfer regardless of material type.



5. Operational Modes

The controller supports at least four operational modes, selected automatically based on the identified material:

CUT mode: Standard cutting operation. Drive frequency and power optimized for the identified material. Used for routine cutting of the expected material type.

CUT-MAX mode: Maximum power cutting. Engaged for dense or hard materials such as cortical bone, hardwood knots, or thick metals. Higher harmonic count and increased power level.

CAUTERIZE mode: Applicable to surgical embodiment. When a blood vessel is identified, the controller switches to a frequency and power combination

optimized for hemostasis rather than cutting.

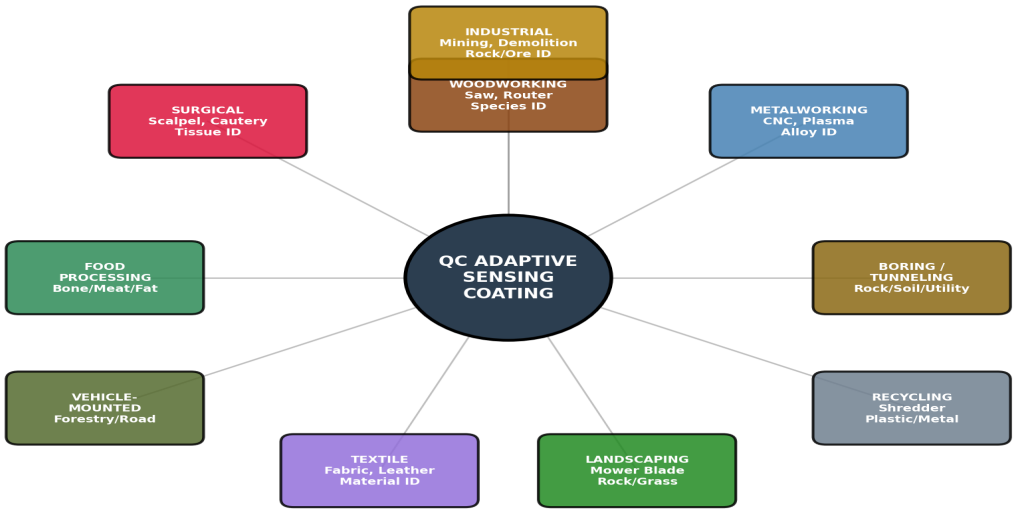
ALERT mode: Safety shutoff or power reduction. Engaged when the identified material indicates a hazard: nerve tissue in surgical applications, embedded nails in woodworking, unexpected material in food processing, or wire/rebar in concrete cutting. The controller may reduce power to zero, activate an audible or visual alarm, or engage a braking mechanism.

IDLE mode: No material contact detected. Drive power reduced to standby level.

6. Application Embodiments

The adaptive cutting system platform supports the following specific embodiments, each using the same core sensing and control architecture with domain-specific calibration of the material lookup table:

FIG. 6 — Application Embodiments: Universal Adaptive Cutting System Platform



Surgical instruments: Scalpels, cautery blades, osteotomes, dermatomes. Material categories: fat (k=0.21), muscle (k=0.49), nerve (k=0.29), blood vessel (k=0.52), cortical bone (k=0.68), tumor (k=0.56), connective tissue (k=0.39). Automatic nerve alert and vessel cauterization. No external sensors required.

Woodworking saws and routers: Circular saw blades, band saw blades, router bits, CNC spindle tools. Material categories: softwoods (pine k=0.12, cedar k=0.11), hardwoods (oak k=0.17, maple k=0.16), engineered wood (plywood k=0.13, MDF k=0.10), metal fasteners (nail k=50-385). Automatic nail detection triggers blade stop within milliseconds. Species identification enables automatic feed rate adjustment.

Metalworking tools: CNC cutters, plasma torch tips, grinding wheels, lathe tools. Material categories: titanium (k=22), steel (k=50), cast iron (k=55), brass (k=109), aluminum (k=205), copper (k=385). Alloy identification from thermal signature enables automatic speed and coolant adjustment. Mixed-material workpiece cutting with real-time adaptation.

Food processing blades: Deboning knives, meat slicers, fish processing blades. Material categories: fat, muscle, bone, cartilage, skin. Automatic bone detection prevents blade damage. Yield optimization by distinguishing meat from connective tissue.

Industrial cutting: Concrete saws, mining tools, demolition equipment. Material categories: concrete (k=1.7), rebar (k=50), rock types (granite k=2.5, sandstone k=1.7, limestone k=1.3), soil (k=0.25-2.0). Rebar detection in concrete, ore type identification in mining.

Textile and leather processing: Fabric cutters, leather splitters, garment industry automated cutting. Material categories: cotton (k=0.04), polyester (k=0.15), leather (k=0.16), silk (k=0.04), metal grommets (k=50+). Automatic material identification for multi-material garments.

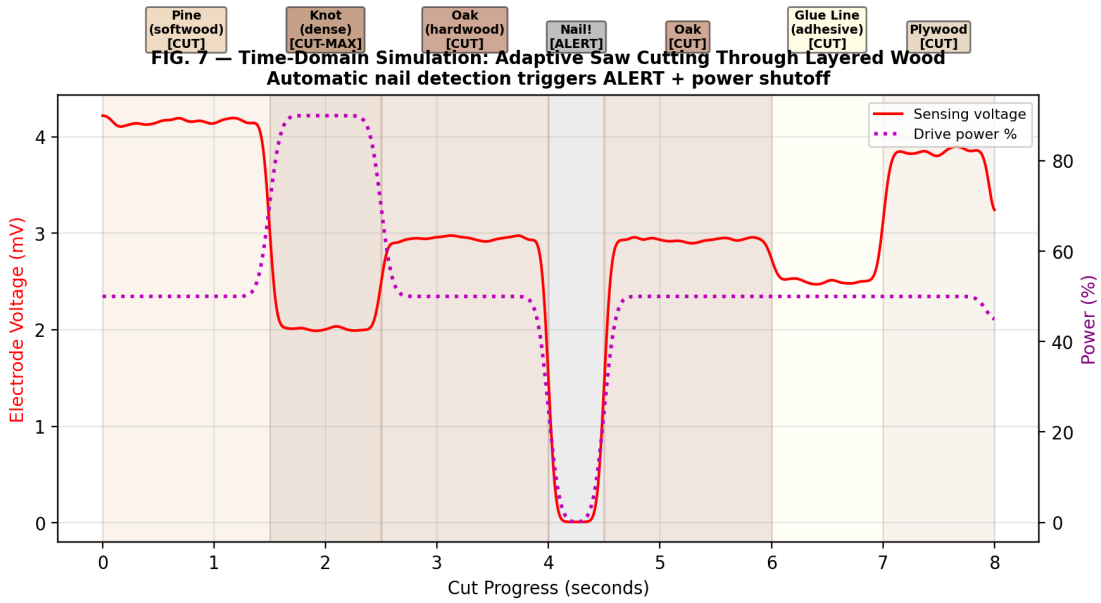
Landscaping equipment: Mower blades, brush cutters, edgers. Material categories: grass (k=0.25), wood (k=0.12-0.17), rock (k=1.5-3.0), metal debris (k=50+). Automatic power reduction when blade contacts rock or metal to prevent damage and projectile hazards.

Recycling and waste processing: Shredders, granulators, size reduction equipment. Material categories: plastic (k=0.15-0.30), paper (k=0.05), metal (k=22-385), glass (k=1.0), wood (k=0.12-0.17). Real-time material stream identification enables sorting confirmation and contamination detection.

Boring and tunneling machines: Tunnel boring machine (TBM) cutter heads, horizontal directional drilling bits, auger teeth, microtunneling tools. Material categories: soil types (clay k=1.1, sand k=0.27, gravel k=1.8), rock types (granite k=2.5, basalt k=1.7, sandstone k=1.7, limestone k=1.3), concrete (k=1.7), rebar and steel pipe (k=50), utility conduit (PVC k=0.16, HDPE k=0.46). Real-time geological formation identification enables automatic cutter head torque and thrust adjustment. Utility strike detection triggers immediate halt to prevent damage to buried infrastructure.

Vehicle-mounted cutting systems: Forestry mulcher heads, road maintenance milling machines, agricultural harvester blades, military engineering vehicle attachments. Material categories vary by application domain. The adaptive coating and controller integrate into vehicle-mounted cutting heads as a subsystem, communicating material identification data to the vehicle control system via CAN bus, Ethernet, or wireless telemetry. Enables autonomous and semi-autonomous operation with real-time material

awareness.



7. Mathematical Model — Validation of Sensing Principle

A mathematical model was developed to validate the thermoelectric sensing principle and adaptive drive selection. The model solves a modified Pennes bioheat equation for the temperature profile through the 13-layer coating stack when contacting materials of varying thermal properties. Key results include:

For biological tissue at 0.5 W ultrasonic power and 1 mm^2 contact area, electrode voltages range from 1.38 mV (blood vessel) to 5.08 mV (fat), with each tissue type occupying a distinct region in 2-channel voltage space. The voltage signal is well within the measurement range of standard 24-bit ADCs (noise floor approximately 0.001 mV at 1 kHz bandwidth).

For frequency coupling analysis, Fibonacci harmonic drive achieves 1.58x better coupling efficiency for cortical bone (hierarchical structure, 7 organizational levels) compared to integer harmonic drive. Integer harmonic drive achieves 2.0x better coupling for fat tissue (uniform structure). The mean advantage of the adaptive selection strategy (choosing the better mode for each material) is 1.3x to 1.6x over fixed single-frequency drive.

Signal-to-noise analysis: at 0.5 W drive power, the minimum electrode voltage (copper contact, k=385) is approximately 0.001 mV, which is at the noise floor of a 24-bit ADC. In practice, materials with k greater than approximately 100 W/m-K would require increased drive power or signal averaging for reliable identification. Materials with k less than 100 W/m-K (encompassing all

biological tissue, wood, plastics, and many metals including titanium and steel) produce voltages well above the noise floor and are reliably identifiable.

CLAIMS

1. An adaptive cutting system comprising: (a) a cutting element having a cutting substrate with a cutting edge; (b) a quasicrystalline coating disposed on said cutting substrate, said coating comprising a plurality of layers arranged in a helical architecture with golden-angle rotational offset between successive layers, wherein at least three of said layers comprise electrically conductive thermoelectric electrode layers positioned at spaced intervals through the coating thickness; (c) a piezoelectric transducer mechanically coupled to said cutting substrate; (d) a controller operatively connected to said electrode layers and said piezoelectric transducer, said controller configured to: (i) measure thermoelectric voltages generated across pairs of said electrode layers when said cutting edge contacts a material; (ii) identify the material in contact with said cutting edge by comparing said measured voltages against a stored library of material signatures; and (iii) select and generate a drive signal for said piezoelectric transducer based on the identified material, said drive signal comprising a frequency pattern, a power level, and an operational mode.

2. The adaptive cutting system of claim 1, wherein said quasicrystalline coating comprises 13 layers each approximately 70 nm thick, with said thermoelectric electrode layers at positions 2, 8, and 12 counting from the substrate.

3. The adaptive cutting system of claim 1, wherein said thermoelectric electrode layers comprise copper nanowire (CuNW) networks having a Seebeck coefficient of at least 50 microvolts per Kelvin.

4. The adaptive cutting system of claim 1, wherein said controller is configured to select between a Fibonacci harmonic frequency pattern comprising frequencies at f_0 multiplied by phi raised to the nth power for n equals 0, 1, 2, 3 and so forth, where phi is the golden ratio, and an integer harmonic frequency pattern comprising frequencies at n times f_0, based on the identified material.

5. The adaptive cutting system of claim 4, wherein said controller selects said Fibonacci harmonic frequency pattern for materials having hierarchical or self-similar internal structure, and selects said integer harmonic frequency pattern for materials having uniform or periodic internal structure.

6. The adaptive cutting system of claim 1, wherein said operational mode is selected from the group consisting of: cut mode, cut-maximum mode, cauterize mode, alert mode, and idle mode.

7. The adaptive cutting system of claim 6, wherein said alert mode comprises at least one of: reducing drive power to zero, activating an alarm, and engaging a braking mechanism, when the identified material indicates a safety hazard.

8. The adaptive cutting system of claim 1, wherein said controller executes said feedback loop in less than one millisecond.

9. The adaptive cutting system of claim 1, wherein said stored library of material signatures is updatable via machine learning during operation of said system.

10. A method of adaptively cutting a material, comprising the steps of: (a) providing a cutting system having a cutting element with a quasicrystalline coating and at least three embedded thermoelectric electrode layers; (b) contacting said material with a cutting edge of said cutting element; (c) measuring thermoelectric voltages generated across pairs of said electrode layers due to a temperature gradient induced by said contact; (d) identifying said material by comparing said measured voltages against stored material signatures; (e) selecting a drive frequency pattern and power level based on said identified material; and (f) driving a piezoelectric transducer coupled to said cutting element at said selected frequency pattern and power level.

11. The method of claim 10, further comprising automatically switching between a Fibonacci harmonic drive and an integer harmonic drive based on the hierarchical structure of the identified material.

12. The method of claim 10, further comprising automatically reducing power and generating an alert when the identified material is a predetermined hazard material.

13. The method of claim 10, wherein said identifying step discriminates between materials having thermal conductivities ranging from 0.03 W/m-K to 385 W/m-K.

14. An intelligent surgical cutting system comprising: a surgical blade having a quasicrystalline coating with at least three embedded CuNW thermoelectric electrode layers; a piezoelectric transducer; and a controller configured to identify biological tissue type from thermoelectric voltage signatures and automatically switch between at least a cutting mode, a cauterization mode, and a nerve alert mode, wherein said identification and mode switching occur without external sensors.

15. The surgical cutting system of claim 14, wherein said controller identifies tissue types comprising at least fat, muscle, nerve, blood

vessel, bone, tumor, and connective tissue.

16. The surgical cutting system of claim 14, wherein said controller automatically reduces power and generates an alert upon identifying nerve tissue.

17. An adaptive woodworking cutting system comprising: a saw blade element having a blade substrate with cutting teeth coated with a quasicrystalline coating with embedded thermoelectric electrode layers; a drive motor; and a controller operatively connected to said electrode layers and said drive motor, said controller configured to identify wood species and detect metal fasteners from thermoelectric voltage signatures, and to automatically adjust rotational speed, feed rate, or trigger blade stop upon detecting metal.

18. The adaptive woodworking cutting system of claim 17, wherein said controller distinguishes between softwood, hardwood, engineered wood products, and metal fasteners including nails, screws, and staples.

19. An adaptive metalworking cutting system comprising: a cutting tool element having a substrate coated with a quasicrystalline coating with embedded thermoelectric electrode layers; a drive mechanism; and a controller operatively connected to said electrode layers and said drive mechanism, said controller configured to identify metal alloy type from thermoelectric voltage signatures and automatically adjust cutting speed, feed rate, and coolant flow based on the identified alloy.

20. An adaptive boring or tunneling system comprising: a rotary cutter head having a plurality of cutting elements, at least a portion of said cutting elements coated with a quasicrystalline coating with embedded thermoelectric electrode layers; and a controller operatively connected to said electrode layers and to the cutter head drive, said controller configured to identify geological formation type and detect buried utilities from thermoelectric voltage signatures, and to automatically adjust cutter head torque, thrust, and rotation speed based on the identified formation, and to halt operation upon detecting a utility strike signature.

21. The adaptive boring or tunneling system of claim 20, wherein said controller distinguishes between at least clay, sand, gravel, sandstone, limestone, granite, concrete, steel pipe, and plastic conduit.

22. An adaptive vehicle-mounted cutting system comprising: a cutting head attached to a vehicle, said cutting head having at least one cutting element coated with a quasicrystalline coating with embedded thermoelectric electrode layers; a controller configured to identify material type from thermoelectric voltage signatures; and a communication interface transmitting material identification data to a vehicle control system.

23. A material-sensing quasicrystalline coating for application to a cutting element of a cutting system, said coating comprising: 13 layers arranged with golden-angle rotational offset, each layer approximately 70 nm thick, wherein layers at positions 2, 8, and 12 comprise electrically conductive thermoelectric electrode layers; said coating configured such that when the coated cutting element contacts a material, the thermoelectric voltages across said electrode pairs are diagnostic of the thermal conductivity and internal structure of said material.

ABSTRACT

An adaptive cutting system incorporating a 13-layer quasicrystalline (QC) coating with three embedded copper nanowire (CuNW) thermoelectric electrode layers at positions 2, 8, and 12 within the coating stack. When the cutting edge contacts a material, the temperature gradient across the coating generates Seebeck voltages at the electrode pairs that are diagnostic of the material's thermal conductivity and internal structure. A controller reads these voltages, identifies the material from a stored signature library, and adaptively selects between Fibonacci harmonic drive (for hierarchically structured materials) and integer harmonic drive (for uniformly structured materials), along with appropriate power level and operational mode (cut, cauterize, alert, idle). The platform applies universally to surgical systems, woodworking saws, metalworking tools, food processing blades, industrial cutting equipment, textile cutters, landscaping equipment, and recycling shredders. Mathematical modeling confirms reliable material discrimination across thermal conductivities spanning four orders of magnitude (0.03 to 385 W/m-K) and demonstrates 1.3x to 1.6x improved coupling efficiency from adaptive drive selection compared to fixed single-frequency operation.

DRAWINGS

FIG. 1 — Cross-Section of 13-Layer Quasicrystalline Coating with Embedded CuNW Thermoelectric Electrodes (Layers 2, 8, 12)

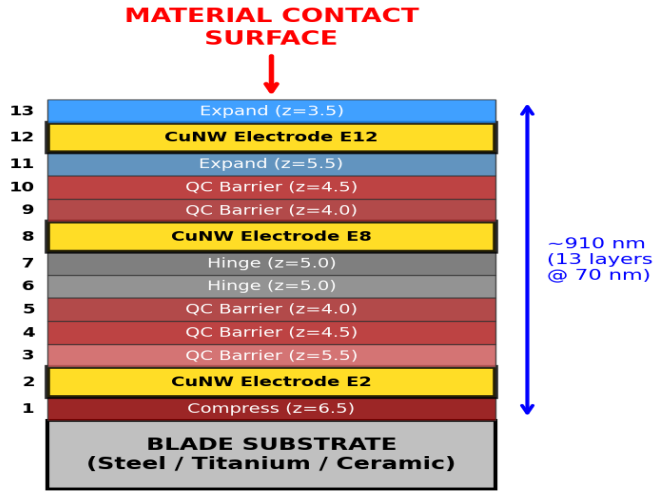


FIG. 2 — Closed-Loop Adaptive Feedback Control System

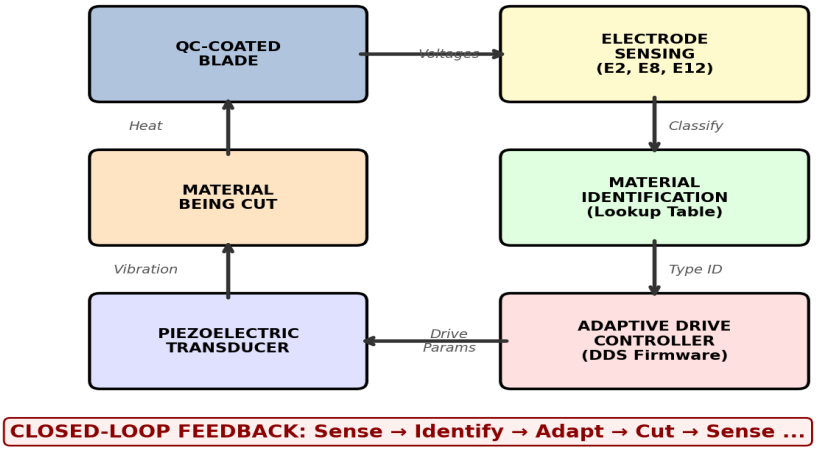


FIG. 3 — Thermoelectric Signatures Across Material Categories
(Voltage inversely proportional to thermal conductivity)

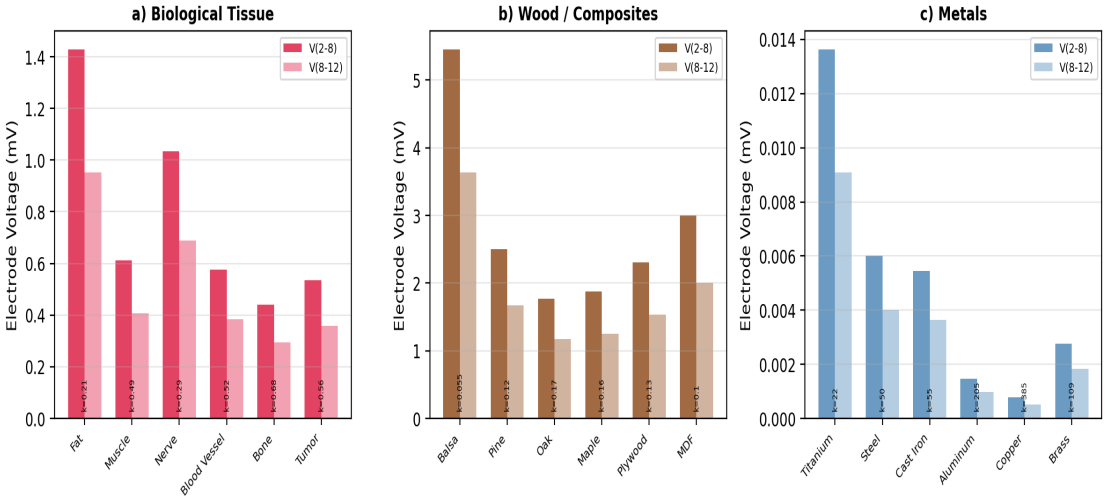


FIG. 4 — Golden Angle vs Equal Spacing Layer Orientation

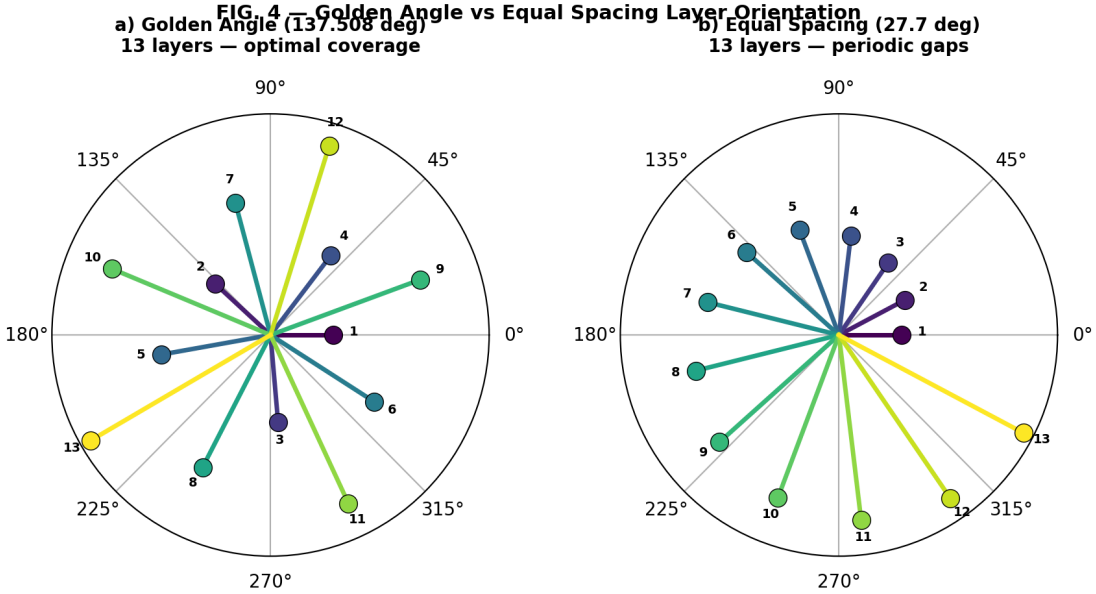


FIG. 5 — Fibonacci vs Integer Harmonic Coupling Across Material Types
Controller adaptively selects optimal drive mode

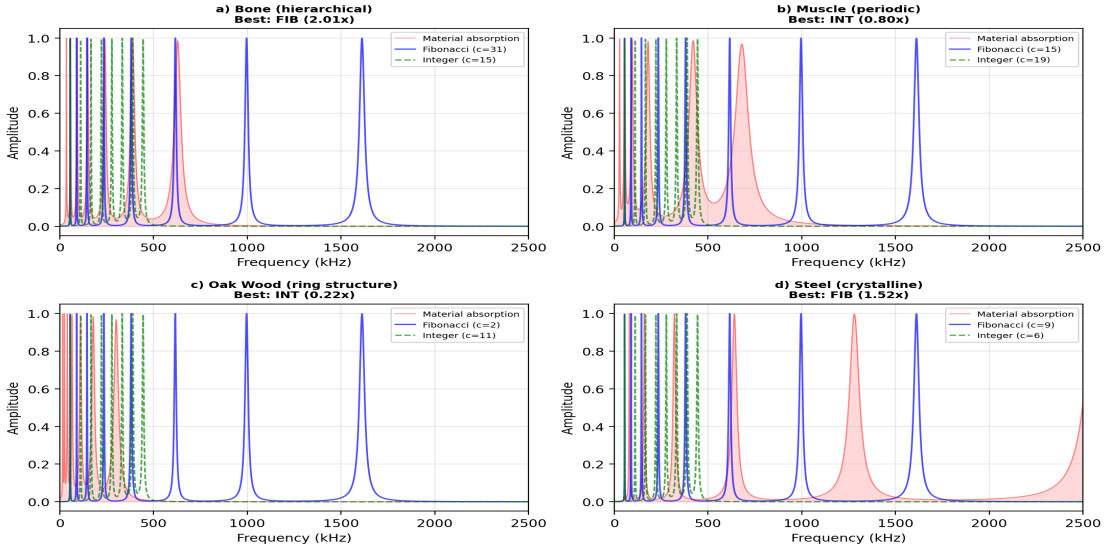
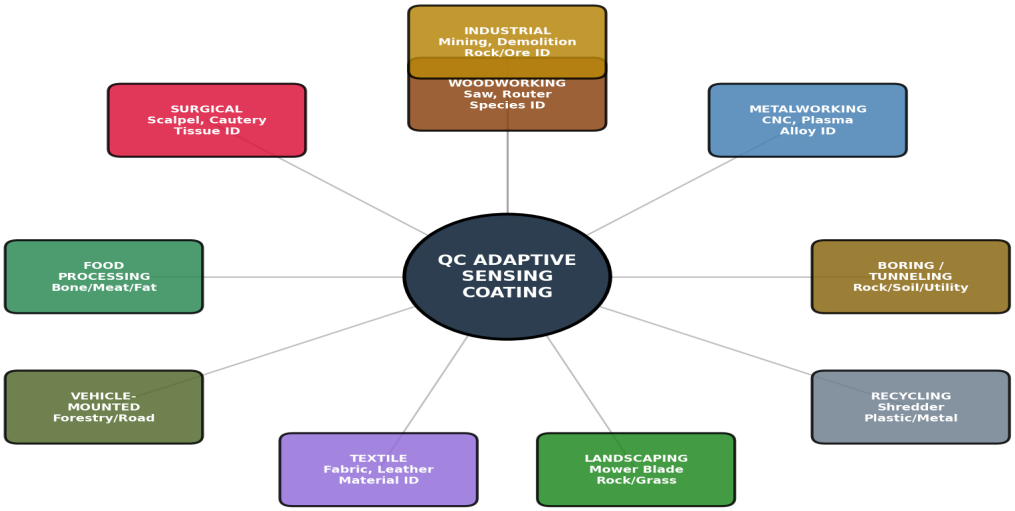


FIG. 6 — Application Embodiments: Universal Adaptive Cutting System Platform



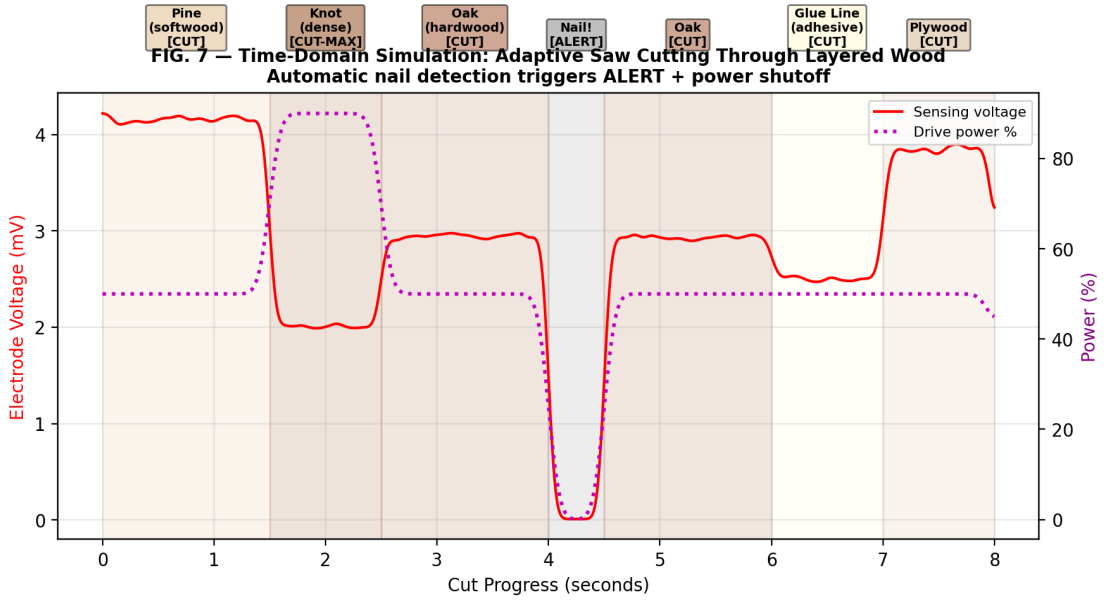


FIG. 7 — Time-Domain Simulation: Adaptive Saw Cutting Through Layered Wood
Automatic nail detection triggers ALERT + power shutoff